

$$\left| \frac{a_{n+1}}{a_n} \right|$$

$$(a_n)$$

$$\frac{n+1}{n} \rightarrow 1$$

$$\ln n \rightarrow \frac{\ln(n+1)}{\ln n} \rightarrow 1$$

Q24

$$\sum \frac{5n^3 - 3n}{n^2(n-2)(n+5)}$$

$$5n^3 - 3n \sim \underline{5n^3}$$

$$n^2(n-2)(n+5)$$

$$\sim n^2(n)(n) = \underline{n^4}$$

$$g(n) = \frac{5n^3}{n^4} = \boxed{\frac{5}{n}}$$

Q28 $\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^3}$

DCT: $\text{Const} \ll \underline{\ln n \ll n^c}$

Does not work $\begin{cases} \ln n \leq n \\ (\ln n)^2 \leq n^2 \end{cases} \Rightarrow \frac{(\ln n)^2}{n^3} \leq \frac{n^2}{n^3} = \frac{1}{n}$

$\ln n \leq n^{1/2}$ for $\underline{n \geq 1}$ Check yourself

$(\ln n)^2 \leq n$
 $\Rightarrow \frac{(\ln n)^2}{n^3} \leq \frac{n}{n^3} = \frac{1}{n^2}$

Joke: $\sin x = x$

$$x = \frac{1}{n} \quad \begin{matrix} x \rightarrow 0 \\ n \rightarrow \infty \end{matrix}$$

Formal: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \Rightarrow \lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n}}{\frac{1}{n}} = 1$

Q45 $\sum \sin \frac{1}{n}$

Q46 $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$

" $\lim_{x \rightarrow 0} \frac{\sin x}{\cos x} \cdot \frac{1}{x}$

" $\lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{1}{\cos x}$



9.5 Q38

$$\sum \frac{n!}{(-n)^n}$$

Root: $\sqrt[n]{\left| \frac{n!}{(-n)^n} \right|} = \frac{\sqrt[n]{n!}}{n} \rightarrow \text{Problem}$

Ratio: $\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!}{n!} \cdot \frac{n^n}{(n+1)^{n+1}}$
 $= n \cdot \frac{n^n}{(n+1)^{n+1}}$
 $= \frac{n^{n+1}}{(n+1)^{n+1}} = \left(\frac{n}{n+1} \right)^{n+1}$

$$\lim \left(\frac{n}{n+1} \right)^{n+1}$$

$$\left(1 + \frac{x}{n} \right)^n \rightarrow e^x$$

$$= \lim \left(\frac{1}{1 + \frac{1}{n}} \right)^{n+1}$$

$$= \lim \underbrace{\left(\frac{1}{1 + \frac{1}{n}} \right)^n}_{\frac{1}{e}} \underbrace{\left(\frac{1}{1 + \frac{1}{n}} \right)}_1 = \frac{1}{e}$$